Predicting the Next Category and Region of a Visit: A Check-in-Level Multi-Task Study on Mobility Data

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***Abstract*—Location-based social networks (LBSNs) record where people go and what they do, one check-in at a time. If we can anticipate the next visit, mobile and urban services can act ahead of demand. Two coarse questions about the next visit are usually enough: its category (the kind of place) and its region (which part of the city). These are normally handled by separate models. We therefore ask whether one model can learn both, and what sharing a single model costs. We first build a check-in-level representation that describes each visit in its own context, instead of giving every place one fixed vector. Across six datasets, five U.S. states and one non-U.S. city (Istanbul), this lifts next-category prediction by a wide margin over a standard place embedding (about $+28$ to $+40$ macro-averaged F1), and most of the gain comes from the per-visit context. We then train one multi-task model for both tasks. At every dataset it outperforms a dedicated category model (about $+5$ to $+9$ macro-F1), and on region it outperforms the dedicated region model at four of the six datasets, while matching it (statistically, within two points) at the other two. Across the five U.S. states the gain on region grows with the number of regions, with the two largest states measured at a single seed. On the non-U.S. city the result holds: the joint model outperforms on category and comes out slightly ahead on region.**

***Index Terms*—next-category prediction, next-region prediction, multi-task learning, check-in-level representation, location-based social networks, mobility data**

I. Introduction

Location-based social networks, such as Gowalla [1], let people share the places they visit. Each check-in is a small record of human movement, and together these records show how people move through a city. A natural and useful question is what a person will do next. Individual mobility is highly predictable in principle [2], and a service that can anticipate the next move can prepare ahead of time instead of reacting after the fact. For a mobility-aware service this can mean staging the right content in the area a user is heading to [3], or planning capacity there before demand arrives. The same logic is already established at the network level, where predicted handovers let cellular services adapt in advance [4]; at the city scale, check-in mobility analysis is likewise read from past traces and treated as a design input for mobility-aware services [5]. We ask the predictive, city-scale version of the question.

Predicting the next place exactly, a single point of interest (POI), is hard, and it is often more than a service needs. Two coarser (less detailed) questions are easier to answer and usually enough: the next category, the kind of place such as food or shopping, and the next region, the part of the city a user is heading to (a neighborhood-scale unit, the census tract in the U.S. and the *mahalle* in Istanbul). These two questions pair naturally, one for intent and one for geography. The question we study is whether one model should learn both at once.

Sharing one representation across tasks is not free. Joint training can help one task while hurting the other, a well-known trade-off in multi-task learning (MTL): one model does several jobs at once by sharing most of its parts, and as a result the shared parameters settle on a compromise that suits neither task perfectly [6]. Our earlier work [7] observed exactly this for next-category and next-region. So the useful question is not whether MTL is good in general, but where sharing helps, where it costs, and how to share so the gains hold and the cost stays small.

We introduce two changes and measure each one carefully. First, we build a check-in-level representation: instead of one fixed vector per place, where two visits to the same coffee shop look identical, each check-in gets its own vector that carries its context (the time, the nearby places, and the recent trail). This extends hierarchical graph representations of places [8], and the infomax idea behind them (training each vector to recognize its true graph neighborhood) [9], by adding a fourth, check-in level beneath the place, region, and city levels (Fig. 1). Per-visit context is not new on its own; what is new is obtaining it from inside this hierarchical graph representation, a choice we test directly (Section VI-A). Second, we train one model that predicts the next category and the next region in a single forward pass, with a shared trunk (a cross-attention stack where the two tasks exchange semantic context) and a private spatial path for the region task (Fig. 2). We deliberately evaluate on two very different settings: five U.S. states of different sizes (Gowalla) and one international city (Istanbul). This lets us see whether the findings hold across small and large, U.S. and non-U.S. data.

Our contributions are the following.

- A check-in-level representation that extends hierarchical graph infomax from the place to the individual visit. It improves next-category prediction by a large and consistent margin over a standard place embedding, about $+28$

to $+40$ points of macro-F1 (a balanced score that counts every category equally) across all six datasets (Table II). A controlled test attributes most of the gain to the per-visit context, not to extra training signal. The margin is large because a single fixed vector per place cannot separate places that host several kinds of activity, which per-visit context recovers.

- A single model for joint next-category and next-region prediction that shares a semantic context across the two tasks while keeping a private path for the spatial one. In one forward pass it outperforms a dedicated single-task category model at every dataset (by about $+5$ to $+9$ points of macro-F1), even though each dedicated category model is tuned per dataset. On region, it outperforms at four of the six datasets and stays statistically non-inferior within a two-point margin (TOST, a test of equivalence) at the remaining two (Table III).
- An empirical account, across five states of different sizes and one international city, of how the joint gain scales with the number of regions. The category gain holds everywhere, and across the U.S. states the region result moves monotonically, from a non-inferior match (TOST, ± 2 percentage points, or pp) at the small region counts to outperforming the dedicated model at the large, with the largest state (California) showing the largest region gain (Fig. 4); Istanbul, the smallest dataset, also comes out ahead on region. Four of the six datasets are measured at four seeds over five folds on both arms (the joint model and the dedicated one); California and Texas are single-seed and provisional.

The rest of the paper reviews background and related work (Section II), defines the problem and the two tasks (Section III), presents our method (Section IV) and experimental setup (Section V), reports results (Section VI), discusses findings and limitations (Section VII), and concludes (Section VIII).

II. Background and Related Work

A. From place embeddings to per-visit embeddings

A place embedding turns a POI into a vector, so that a model can reason about places by their learned geometry, where similar or nearby places sit close together. This replaces treating each place as a bare, arbitrary index. Deep Graph Infomax (DGI) [9] is a general self-supervised method for learning node representations on graphs. Applied here to a network of places, it learns such vectors by training them to tell the real place network, linked by similarity, time, and distance, from a shuffled copy. Hierarchical Graph Infomax (HGI) [8] adds geographic structure, organizing the network as place, then region, then city, so the vectors respect where places sit. Both methods produce one vector per place, so two visits to the same coffee shop, one at 8 a.m. and one at 10 p.m., look identical to the model. Contextual check-in representations instead give each visit its own vector: CTLE [10] is the closest example, learning a vector per visit by hiding parts of a user's check-in sequence and learning to fill them back in. Our representation, Check2HGI, also works at the check-in level, but it gets there differently. It keeps the hierarchical place-to-region-to-city network and trains it with the same infomax objective, now extended one level deeper, from individual places down to individual check-ins, a fourth level beneath place, region, and city (Fig. 1). It does not switch to a sequence model, as CTLE does. In short, earlier place embeddings (DGI, HGI) changed *which* features a place vector encodes; CTLE and we change *how often* a vector is produced, once per place or once per visit.

The novelty is this specific combination: per-visit context inside a hierarchical graph-infomax representation. We compare against CTLE directly to show that the combination, not per-visit context on its own, is the source of the gain.

B. Predicting the next category and the next region

A large body of work predicts the next place a user will visit, the exact POI, with recurrent and attention models such as ST-RNN [11], DeepMove [12], Flashback [13], STAN [14], and GETNext [15]. In contrast, we target two properties of the next visit, its category and its region, rather than the exact place. These targets are not trivial: the region alone spans hundreds to several thousand classes. We choose them because they are what a mobility-aware service can act on, not to make the task easier. Predicting over a partition of the map is also the standard formulation in the human-mobility literature, where the target is a grid cell rather than an exact place [16]; our next-region task follows this formulation, with administrative units in place of grid cells. The category task itself follows our earlier line of work [7]. The field increasingly models several granularities at once, predicting category-level and region-level sequences alongside the place sequence; in that work, however, category and region are auxiliary signals that help a headline next-place task (SGRec [17], MCMG [18], HMT-GRN [19]). We study the pair as the object itself. To our knowledge, fine-grained region as a co-equal end target, rather than an auxiliary coarse cell, is underexplored: HMT-GRN, for instance, predicts a coarse geohash cell only as an aid to its next-place search. The nearest exceptions stop short of our pairing in one way or another: DRRGNN [20] forecasts a person's next activity region jointly with a mobility-intention label, but over regions discovered per person rather than a fixed citywide partition. A recent generative recommender [21] predicts category and region jointly, but only as auxiliary steps toward its next-place ranking.

C. Multi-task learning for mobility

Multi-task models for mobility have paired location prediction with related signals for some time. MCARNN [22] jointly predicts activity and location, iMTL [23] couples them in a shared, interactive framework built for sparse check-ins, and a recent hierarchy-aware model continues the pattern [24]; in all three the location target is the exact place. The closest alternative to our setting is the cascade, a pattern with a long lineage: predict the category first, then predict the place given the category [25], [26]. CSLSL [27] is its current form, predicting in a chain (when, then what, then where), with location as the headline output and category as an instrumental step along the way; CatDM [28] likewise uses category instrumentally, to filter candidate places. We differ in two ways. We predict category and region in parallel, as co-equal end targets in one model with a single forward pass, and we drop the next-place target entirely, so neither task is instrumental to a third. The parallel form also follows from the task pair itself: with two co-equal targets, a chain would have to promote one of them to an upstream stage whose errors the other inherits, an order nothing in the tasks dictates. Since the cascade is the pattern the field actually uses, we also test the choice directly (Section VI-B). On the optimization side we are deliberately conservative: training one model jointly on several tasks with a fixed loss weighting is standard practice [6], and our use of it is not itself the contribution. Rather than propose yet another gradient-balancing method, a training scheme that re-weights the tasks' updates on the fly (PCGrad [29], GradNorm [30], Nash-MTL [31]), we ask where sharing helps and where it costs, and we give a clear, measured account, consistent with reports that such methods rarely improve on a well-tuned fixed weighting at two tasks [32], [33]. We confirm this on our own two tasks: across the gradient-balancing methods we tried, including the three named above, none improved on a tuned fixed task weighting in our model. The reason is visible in the gradients themselves: the next-category and next-region updates barely correlate on the shared layers (near zero throughout training), so the two tasks neither pull against each other nor reinforce each other there, and a balancer has no conflict to resolve. This is consistent with the mechanism test we report in Section VI-B. We report this as our finding for this pair of tasks, not as a general rule. The novelty here is empirical, not architectural.

III. Problem and Tasks

We work with check-in sequences. For each user we order their check-ins in time and form short windows of nine consecutive visits. From each window we predict two properties of the next visit. The first is its *category*, the kind of place, one of seven fixed labels: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. Istanbul's places are mapped onto the same seven labels by its source collection (Section V-A). The second is its *region*, the census tract the place falls in (the *mahalle* for Istanbul). We do not predict the exact next place; these two properties are easier to learn and, for most uses, enough. Category is a small, fixed label set. Region is a large one: we frame next-region as classification, where each candidate census tract is one class, from about five hundred (Istanbul) to about eight thousand five hundred (California).

The motivation is practical: the category says what kind of place comes next, which indicates what a user will want; the region says where, which indicates where to prepare. Such preparation has support in the mobility literature, which has long built city services on location-based social network traces [34]. A recent analysis of tourist check-in mobility finds that visits concentrate on a few hub places, argues that crowds and demand therefore concentrate as well, and ties this structure to infrastructure planning and adaptive strategies [5]. That analysis reads the structure of past visits; we predict two properties of the next one, and aggregated over users those predictions point to where demand is heading before it lands. A census tract is a neighborhood, not a radio cell (the small coverage area a single base station serves), so we scope our claims to neighborhood-level preparation: demand and load anticipation, content staging, and capacity planning; not cell association or handover. We build and evaluate no such service here; Section VII quantifies what these predictions would give such a service.

IV. Method

A. The check-in-level representation

We want each visit, not each place, to carry its own vector. A standard place embedding gives every point of interest one fixed vector, so two visits to the same café look identical even when they happen at different times of day, in different company, or after different trips. We instead describe each check-in in its own context. Figure 1 shows the full data flow, from the raw check-in trail to the two outputs.

We build a graph with four levels: the check-in, the place it records, the region (a census tract) the place sits in, and the city. Edges tie these levels together and tie check-ins to one another as consecutive visits by the same user, weighted by how close in time they fall; visits to the same place are tied through their shared place node one level up, and nearby places are linked at the place level. Each visit's category, time of day, and day of week enter as input features of its node rather than as edges. This keeps the place-to-region-to-city geography of a hierarchical graph and adds the check-in as a new bottom level. We train the graph mainly with an infomax objective, which contrasts real neighborhoods against shuffled ones, so each vector learns to match its true neighborhood and reject a fake one [8], [9], plus two small label-free auxiliary terms: a masked reconstruction of each place's aggregated category features and an anchor to a place embedding pre-trained, label-free, on the same data (weights 0.3 and 0.1). The training uses no task label: it never sees the next category or the next region. From the trained graph we read out one 64-dimensional vector per check-in (its region nodes carry trained vectors as well), and a model then sees a sequence of these per-visit vectors rather than a sequence of repeated per-place ones.

Each ingredient is individually standard: per-visit context, hierarchical place graphs, and the infomax objective all appear in prior work; as Section II-A argued, the contribution is their combination. We show later that this combination, not extra supervision, is what makes the category task far easier to learn.

B. The single multi-task model

We then train one model that reads a window of recent per-visit vectors and predicts two properties of the next visit at

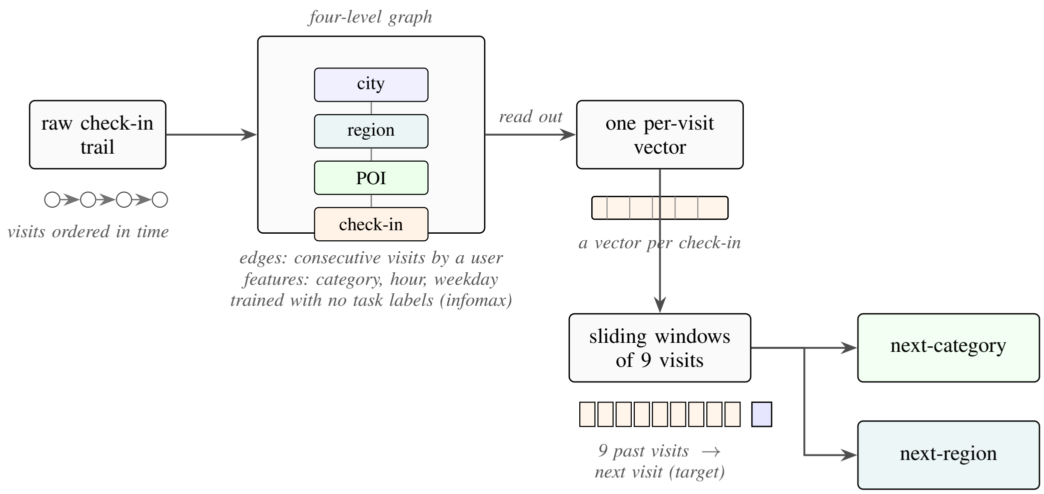

Fig. 1. From a check-in trail to two predictions: each visit is embedded in a four-level graph (check-in, place, region, city) and described in its own context; overlapping nine-visit windows feed a single model that outputs the next category and the next region.

once: its category and its region. Figure 2 shows the design. Two input streams, a semantic stream with the window of per-visit vectors and a spatial stream in which each visit is represented by the trained vector of its region node from the same graph, pass through private per-task encoders (a small input network dedicated to each task, with no weights shared between them) into a shared cross-attention stack of two blocks, where the two streams exchange information: in each block, attention lets each stream read the other's features, while each stream keeps its own feed-forward weights. The tasks therefore share by exchanging information between per-task streams, not by owning hidden layers in common. The stack then feeds two outputs, one for the category and one for the region. The region output keeps a private spatial path of its own, a small branch that the category task does not touch.

The private spatial path is a task-specific branch inside the one model, not a second model. The whole system is a single model: one forward pass produces both predictions, the deployment property we care about.

The training loss is a fixed-weight sum of the two outputs, and both outputs use plain unweighted cross-entropy. One might expect the class-balanced category metric to favor a class-weighted loss; we tested class-weighting on both heads and it lowered both region accuracy and category macro-F1. We therefore keep plain unweighted cross-entropy on both and balance the two tasks with a fixed-weight sum, the category output weighted 0.75 and the region output 0.25. This is ordinary fixed-weight joint training [6], kept deliberately plain. Any improvement of the joint model over the dedicated single-task models then comes from the shared representation, not from an adaptive weighting scheme.

The cost of serving both tasks this way is not free, and we state it plainly. The joint model carries the cross-attention stack and both task heads, so it is larger than either dedicated single-task model of Table III, and a forward pass costs more compute than running the two small dedicated models: about 4.2 million parameters at Alabama against 1.1 million for the two combined (5.2 against 2.0 at California). What the single model buys is operational rather than arithmetic: one artifact to train, version, and deploy, and one forward pass whose one set of inputs produces both answers at once.

V. Experimental Setup

A. Data

We evaluate on six datasets, summarized in Table I. Five come from Gowalla, a location-based social network (collected 2009 to 2011), and cover the U.S. states of Alabama, Arizona, Florida, Texas, and California [1]; the sixth is the city of Istanbul, drawn from the Massive-STEPS collection [35], which gives us a non-U.S. check on the findings. The states run from small to large, from about 114 thousand check-ins and 1,109 regions in Alabama to several million check-ins and 8,501 regions in California, so we can see whether a result holds as the number of regions grows. Every dataset uses the same seven place categories: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. The region unit is a census tract for the Gowalla states and a mahalle (a municipal neighborhood) for Istanbul, and the number of regions ranges from about five hundred to about eight thousand five hundred.

B. Windows, splitting, and the integrity of the representation

Windows. For each user with at least ten visits we sort the check-ins in time and form overlapping windows of nine visits, one starting at each visit, with the next visit as the target. Overlapping windows give both tasks more examples to learn from. Because overlapping windows would end many of a user's windows on the same final visit, we keep a single full-context window ending on each user's last visit and drop the padded duplicates.

Splitting. We split by user with stratified five-fold cross-validation, so all of a user's windows fall in the same fold.

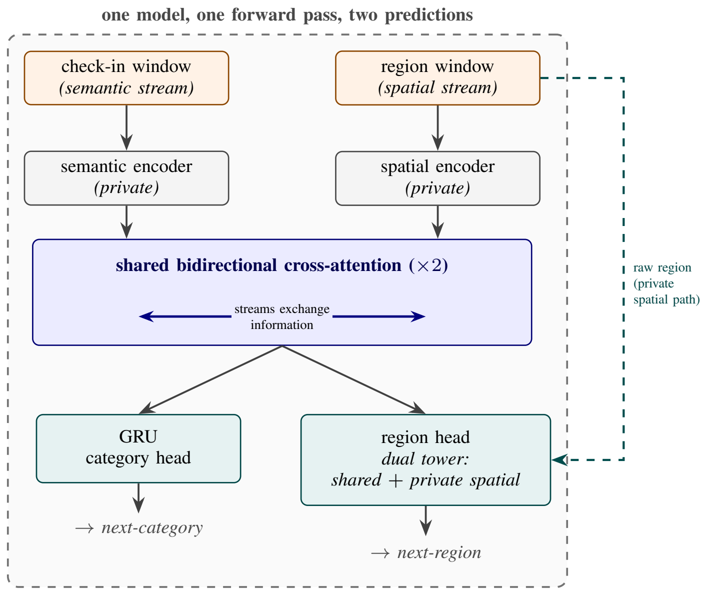

Fig. 2. The single multi-task model. A check-in (semantic) window and a region (spatial) window pass through private encoders into a shared cross-attention stack, the only place the two tasks interact; the category head predicts the next category, and the region head combines the shared output with a private spatial path. One model, one forward pass, two predictions.

Overlapping windows therefore cannot leak across the split, because a test user's visits never appear in training. The split is stratified on the next-category label, which roughly preserves the class proportions across folds, and this matters because the seven-class category task is imbalanced: Food is the majority class in every dataset, from about 25 percent of visits in Florida to 34 percent in Alabama.

Integrity of the representation. We train the representation once on the whole dataset and then feed it to both the dedicated single-task models and the joint model. We argue, and verify, that it carries no usable information about the test visits, on three grounds. First, the representation is trained without any task label: its objective is label-free, contrasting real graph neighborhoods against shuffled ones with small auxiliary terms that reconstruct quantities computed from the inputs themselves, and it never sees the next-category or next-region target, so no label can travel from test to train through it. Second, the only thing it could carry from the test side is graph structure (which places sit near which), because the graph is built over all places. We measured that exposure directly, rebuilding the representation per fold from that fold's training users only and re-running both tasks against the full-corpus version on the same folds. The effect is within fold noise, with no systematic gain: on region the change is -0.33 points at Alabama, $+0.01$ at Arizona, and -0.12 at Florida. On category it is $+0.29$ at Alabama, $+0.27$ at Arizona, and $+0.00$ at Florida, measured on the in-coverage visits whose places appear in training (about 67 percent at Alabama to 87 percent at Florida); this is a place-level proxy, since the representation cannot score a place never seen in training. Third, the one component that could pass visit-to-visit information is a region-transition prior: a table of how often visits move from one region to the next, used to bias region scores. An earlier version that used the whole dataset inflated region accuracy by 13 to 27 points across states, which is why any such table is built per fold, from training data only, seed by seed. The joint and dedicated models reported here do not use this prior; it survives only in the HMT-GRN baseline (Section V-D), with the per-fold training-only build. The place-level baseline representation (HGI) is pre-trained once on the whole dataset, exactly like ours; the contextual baseline (CTLE, Section V-D) is pre-trained per fold on training users only, a stricter standard, so no baseline input is held to a weaker standard than ours. The one residual we cannot fully measure is visits to places never seen in training: the representation is trained on all places, so it cannot score a place that is absent from training. On the visits we can measure, which are the large majority, the audit finds no inflation.

C. Metrics, superiority versus non-inferiority

For the category task we report macro-averaged F1 (macro-F1), which averages the per-class F1 so that each of the seven categories counts equally, including the rare ones; its reference point is the majority-class floor, the macro-F1 of always predicting the single most common category. For the region task we report accuracy at ten (Acc@10), the fraction of test visits whose true region appears among the model's ten highest-scoring guesses (a visit whose true region is absent from that fold's training data counts as an error); its comparison anchor is the dedicated single-task model.

We make two kinds of comparison and test each one appropriately. Where the joint model is meant to outperform, we test *superiority* with paired tests over matched runs: at the four datasets measured at four seeds, a paired test over the per-seed means with a Holm correction across the cells; at the two single-seed states, a paired Wilcoxon signed-rank test over the folds. Where the joint model is meant only to stay even, we test *non-inferiority*, that it is no worse than the dedicated model by more than a two-point margin, with a two-one-sided-tests (TOST) procedure. We fix the two-point margin in advance and on deployment grounds, before looking at whether any claim survives it: as Section III argued, a mobility-aware service acts on which region will be busy, not on a single rank position, so a two-point shift in Acc@10 is below the granularity at which such a service would behave differently. The margin is also large relative to the task: with five hundred twenty to eight thousand five hundred regions, a random top-ten guess lands the true region at most about two percent of the time, so two points span a wide band of the achievable range. Finally, the equivalence is well powered. The standard deviation of the paired difference is small (0.04 to 0.15 points at the four multi-seed datasets, 0.07 at the two single-seed ones), which gives a power near 1.0 to reject a true two-point gap, so a non-inferiority verdict reflects a real match rather than a noisy test. Where the joint model outperforms and where it matches is a result, reported in Section VI, not here.

D. Baselines

We organize the comparisons into three roles, kept separate so each answers its own question. The first role is the state of the art for each task, on our data, under user-disjoint five-fold protocols. For the next-category task this is POI-RGNN [36], run as a faithful re-implementation, and a simple "what usually follows what" Markov baseline over the recent categories, reporting the strongest order per dataset (Markov-K, up to the nine-visit window). For the next-region task this is HMT-GRN [19], our primary region-native comparison, predicting the region from its own end-to-end recurrent model on the same data, seed, and folds; we keep its shared recurrent multi-task skeleton, add a region-transition prior built per fold from training data, and drop its graph components and its hierarchical beam search, both of which serve the next-place head we do not predict. The HMT-GRN comparison is therefore a region-native model, not a reproduction of the complete published system. Alongside it we run a faithful STAN [14], trained from raw with its own embeddings (not ours) and its own sequence construction, under one audited recipe, and a ReHDM [37] reference reported under that method's own published protocol rather than re-run under our matched protocol. The second role is a pair of representation controls that attribute the category gain to our specific design, not to contextualization in general or to extra features: CTLE [10], the closest prior contextual check-in embedding, run end to end at Florida and, in frozen form, at Alabama, Arizona, and Istanbul; and a feature-concatenation control that appends raw per-visit features to a standard place embedding under the same model. The third role is a design comparison between the two ways of coupling the tasks. The dominant published coupling is the directed cascade, which predicts the category first and conditions the place prediction on it [25], [26], [28]; CSLSL [27], its current form, even reports the chain outperforming a parallel variant that shares one trunk, so the choice of coupling is not settled in advance. We test the cascade inside our own model rather than through a re-implementation of those systems: we remove the cross-attention stack where the two streams exchange information and instead feed the predicted category forward into the region pathway, keeping the representation, the outputs, and the training recipe unchanged. This isolates what we actually choose, the coupling topology (chain versus parallel), at no additional parameter or compute cost: the chain form runs without the exchange stack, and the added conditioning projection is about a thousand parameters. The dedicated single-task models are the comparison anchors for the joint model throughout, and the standard place-level embedding [8] is the place-level comparison that isolates what the per-visit representation adds. The full training recipe for every model here (optimizer, learning rates, schedule, epochs, batch sizes, seeds, hardware) ships with the released code (Section VIII).

VI. Results

A. The representation makes the category task learnable

Read this as: describing each visit in its own context, rather than each place once, is what makes the next-category task learnable. Table II compares our check-in-level representation against a standard place embedding (HGI [8]) on next-category macro-F1, both fed to the same dedicated model under the windowing used throughout the paper. This is a controlled, like-for-like comparison: the target, the single-task head, the training recipe, and the folds are identical across the two columns, and only the input representation changes (a vector per visit versus a vector per place), so the margin isolates the representation, not a difference in task or model. (The check-in-level column therefore keeps one fixed recipe; it is intentionally not the per-dataset-tuned ceiling of Table III.) The check-in-level representation outperforms the place embedding by a wide and consistent margin at every state: $+29.31$ at Alabama, $+27.63$ at Arizona, $+39.62$ at Florida, $+37.95$ at California, and $+37.47$ at Texas. The same holds on Istanbul ($+28.09$). The gains fall in two bands, about $+28$ to $+29$ points at the smaller datasets (Istanbul, Alabama, Arizona) and about $+37$ to $+40$ at the three large states. In absolute terms the place embedding recovers only about half of what the per-visit representation reaches. A controlled test attributes most of that gain to the per-visit context itself: replacing each visit's vector with a per-place-averaged version of the same representation gives back only part of the margin. This leaves roughly 64 to 90 percent of the gain (state-dependent) to the context each visit carries, not to extra training signal.

Figure 3 shows why the representation helps this task in particular. By category, the per-visit vectors are cleanly grouped. Their silhouette score by category, a separation measure from -1 to 1 where higher means the seven labeled groups are tighter and better separated, is about 0.57, against

TABLE I
DATASET STATISTICS, ORDERED BY NUMBER OF REGIONS: FIVE GOWALLA STATES AND ONE INTERNATIONAL CITY (ISTANBUL, FROM MASSIVE-STEPS). ALL DATASETS SHARE THE SAME SEVEN ROOT CATEGORIES; THE REGION UNIT IS A CENSUS TRACT (A MAHALLE FOR ISTANBUL). WINDOWS ARE THE OVERLAPPING NINE-VISIT INPUTS PLUS THE NEXT VISIT AS TARGET; SEQUENCE LENGTHS ARE PER-USER CHECK-IN COUNTS; DENSITY IS THE CHECK-IN-MATRIX FILL, $100 \times$ CHECK-INS/(USERS $\times$ POIS); MAJORITY IS THE SHARE OF NEXT-VISIT LABELS IN THE MOST COMMON CATEGORY (FOOD IN EVERY DATASET).

Dataset	Source	Check-ins	Users	POIs	Regions	Windows	Max len	Avg len	Density (%)	Majority (%)
Istanbul	Massive-STEPS	462,615	23,694	29,816	520	271,666	817	19.5	0.065	33.4
AL	Gowalla	113,846	3,858	11,848	1,109	96,326	3,835	29.5	0.249	34.2
AZ	Gowalla	236,450	7,869	20,666	1,547	200,895	5,589	30.1	0.145	34.0
FL	Gowalla	1,407,034	21,052	76,544	4,703	1,274,418	16,679	66.8	0.087	24.7
TX	Gowalla	4,089,892	38,644	160,938	6,553	3,830,414	42,300	105.8	0.066	31.0
CA	Gowalla	3,171,380	37,090	169,145	8,501	2,925,466	14,855	85.5	0.051	32.7

about 0.00 for the place embedding. Their nearest-neighbor category purity, the share of a vector's nearest neighbors that carry its own category, is about 0.98, against about 0.78 (both averaged over the five U.S. states). The same geometry does not separate regions, which is exactly the point: the per-visit vectors improve category separability without grouping by region, so their benefit is category-only (the joint model's spatial stream instead reads the region-level vectors of the same graph, Section IV-B), and this sets up the joint study in Section VI-B. A direct comparison with the closest prior contextual embedding makes the source of the gain concrete. CTLE [10] also gives each visit its own vector, but it learns from place identifiers and timestamps alone: its pretraining recovers masked places and masked hours, so the category vocabulary never enters its training, whereas our graph reads each visit's category and time of day as input features (and even the place embedding builds its place vectors from category-derived features). Fine-tuned end to end at Florida at its authors' defaults, scaled to the same 64 dimensions as our representation and under the same windowing, CTLE reaches 33.45 macro-F1 at its best epoch (29.69 at its final one), about two points below the place embedding under the same best-epoch rule and far below our 75.15; frozen and fed to the same single-task model as ours, the control at Alabama, Arizona, and Istanbul repeats the ordering, with margins of $+37.8$, $+37.0$, and $+28.7$ macro-F1. A feature-concat control, the place embedding joined with raw per-visit features and fed to the same model, does not close the gap either, so the category gain comes from the hierarchical per-visit representation, not from contextualization alone and not from feature injection.

B. One model, two tasks

Read this as: one model outperforms a dedicated next-category model at every dataset and outperforms or matches a dedicated next-region model, with the region gain rising with the number of regions across the U.S. states. To calibrate the two scales: category macro-F1 is taken over seven classes (a majority-class floor of about 7%), and region Acc@10 is taken over hundreds to thousands of regions, where a random top-ten guess is right at most about two percent of the time (and well under one percent at the larger region counts). Table III reports the single joint model against the dedicated single-task ceilings (we call them ceilings because a jointly trained head is usually expected to match or fall short of a head trained for that task alone). Each dedicated category ceiling is tuned per dataset over batch size and learning rate (the region ceilings use the strongest fixed recipe), so the joint model is compared against the strongest dedicated setting, not a convenient one. Figure 4 plots the signed gains ordered by region count. On category the joint model outperforms the dedicated ceiling at every dataset, by $+5.34$ to $+9.40$ macro-F1 (smallest at Florida, largest at Arizona, and $+8.59$ at Istanbul). On region, measured by Acc@10, the joint model outperforms the dedicated ceiling at Florida $+0.72$ (4,703 regions), Texas $+2.12$ (6,553), California $+2.20$ (8,501), and Istanbul $+0.28$ (520), and stays a non-inferior match (TOST, ± 2 pp) at Alabama (-0.31) and Arizona ($+0.10$). Across the five U.S. states the region gain rises with the number of regions, from -0.31 at the smallest to $+2.20$ at the largest, so the joint model helps the spatial task more, not less, as the number of regions grows, the opposite of a cost that grows with the size of the spatial problem. We order by region count because that is the axis of interest; region count and corpus size co-vary across these datasets, so we read the trend across the points rather than as a precise law.

Four of the six datasets are measured at n=20 on both arms: Alabama, Arizona, Florida, and Istanbul average four seeds over the same five frozen folds, and the tests pair the per-seed means (four paired observations per dataset). On category, all four seeds favor the joint model at every one of these datasets, and each per-dataset gain is significant after a Holm correction across the four cells (paired t, corrected $p < 0.001$ everywhere). On region, all four are tested equivalences (TOST, ± 2 pp), and all four pass with 90% confidence intervals well inside two points: Alabama (-0.31; -0.48 to -0.14), Arizona ($+0.10$; $+0.00$ to $+0.21$), Florida ($+0.72$; $+0.67$ to $+0.78$), and Istanbul ($+0.28$; $+0.24$ to $+0.32$). At Florida and Istanbul the whole interval sits above zero, so the joint model outperforms there as well, though those two gains are smaller than the two-point margin of Section V-C; the region gains above that margin are at Texas and California. At Arizona the interval touches zero, so we report a match, not a gain. At Alabama the interval sits entirely below zero, a small but statistically significant deficit, still well within the two-point margin. California and Texas remain provisional: their dedicated ceilings are n=20, but the joint cells are a single seed over five folds. On those folds, all five favor the joint model on both tasks (one-sided paired Wilcoxon, p=0.031, the smallest value five folds can produce), and the region gains carry 90% intervals entirely above the two-point margin (+2.14 to +2.28 at California, +2.04 to +2.18 at Texas). A multi-seed run at these two states is the remaining confirmation.

A reader used to multi-task learning [6] expects the harder task to pay for the easier one, so it is worth saying where the category gain comes from. A controlled test isolates the source: we freeze the region pathway at the start of training so it cannot learn, and therefore cannot teach the category task, yet the full category lift survives at Alabama, Arizona, and Florida (within 0.3 of the joint model and far above the dedicated single-task model). We therefore read the category gain as a stronger shared trunk, obtained without a second model to serve (one model, one forward pass), rather than the region task teaching the category one; we report this as a finding, not a hypothesis.

The joint model is also above every external baseline reported, on both tasks, at every dataset (Table III): the primary region-native comparison, HMT-GRN [19] (for example California 49.61 versus our 65.69), a from-raw STAN [14] and a ReHDM reference [37] on region, and POI-RGNN [36] and a Markov floor on category, all trail it. These externals run on their own embeddings, so this comparison folds in the representation advantage of Section VI-A; the like-for-like anchor remains the dedicated column. A simple Markov region floor reaches 43 to 65 Acc@10 across the datasets (computed under a non-overlapping windowing of the same data, indicative rather than protocol-matched); the joint model clears it by 12 to 23 points everywhere.

We also test our choice of coupling the tasks in parallel rather than in a chain. We prefer the parallel form on principle: the two targets are co-equal, so a chain would have to pick one to predict first and pass its errors to the other. The empirical record still had to be checked, because the cascade dominates published practice, and CSLSL reports its chain outperforming a parallel variant that shares one trunk on its own benchmarks [27]. Rewired as a cascade (Section V-D), our model reaches the same combined score, the geometric mean of the two task metrics, as its parallel form at no additional parameter or compute cost: the difference is about zero at Alabama, Arizona, Florida, and Istanbul, the four datasets where we ran the comparison, with both forms trained under the same configuration and single seed, and neither task moving by more than a quarter of a point. We read this as a defense of the parallel design, not a claim that we outperform the cascade: on this representation the coupling topology makes no measurable difference in either direction, and the simpler parallel structure loses nothing. Two qualifications bound this reading: the cascade runs under the recipe tuned for the parallel model, and its form was fixed in advance, so a cascade tuned as carefully as the parallel design remains future work.

TABLE II
NEXT-CATEGORY MACRO-F1 OF THE CHECK-IN-LEVEL REPRESENTATION AGAINST A STANDARD PLACE EMBEDDING (HGI), SAME SINGLE-TASK HEAD, RECIPE, AND FOLDS (MEAN AND FOLD STANDARD DEVIATION, SEED 0).

Dataset	Check-in level	Place level	Δ
Istanbul	**54.65** ±0.6	26.56 ±0.5	+28.09
AL	**55.87** ±2.4	26.56 ±1.0	+29.31
AZ	**57.13** ±1.3	29.50 ±1.0	+27.63
FL	**75.15** ±0.5	35.53 ±0.3	+39.62
TX	**69.95** ±0.2	32.48 ±0.6	+37.47
CA	**70.26** ±0.3	32.31 ±0.4	+37.95

The matching place-level value at Alabama and Istanbul (26.56) is a coincidence of two independent runs; their per-fold values differ.

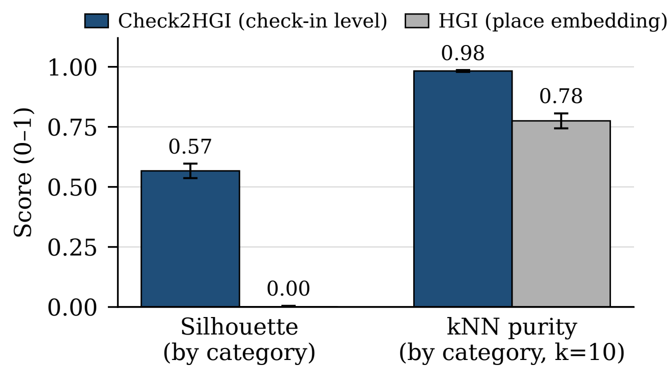

Fig. 3. Class separability of the representations, grouping each vector by its ground-truth category label (cosine silhouette; leave-one-out nearest-neighbor (kNN) purity, k=10), averaged over the five U.S. states. The check-in-level representation separates the seven categories; the place embedding does not.

C. External validity (Istanbul)

Read this as: the finding holds on a non-U.S. city, measured on the same representation and protocol as the U.S. states. On Istanbul the joint model outperforms the dedicated category ceiling by +8.59 macro-F1 (54.74 to 63.33) and edges past the dedicated region ceiling at +0.28 Acc@10 (75.16 to 75.44), a statistically supported gain (the 90% confidence interval of the paired difference lies entirely above zero; it also passes the non-inferiority test, TOST, ±2 pp); both arms average four seeds (n=20). As with every dataset, the comparable quantity is the lift over the dedicated ceiling, not the absolute Acc@10: region counts differ across datasets (520 mahalle here), so absolute scores are not comparable between them. The representation, the overlapping-window protocol, and the training setup are identical to the Gowalla states', so the comparison is like-for-like. The external baselines line up the same way as on the U.S. states: on category the joint model is far above the Markov floor (24.55) and POI-RGNN [36] (30.12), and on region it is above ReHDM [37] (69.33), a faithful STAN [14] (61.86), and HMT-GRN [19] (60.4). The picture from the U.S. states repeats on a different continent and a different region unit, and on region Istanbul comes out slightly ahead rather than merely close.

TABLE III

ONE MODEL, TWO TASKS: THE SINGLE JOINT MODEL AGAINST THE DEDICATED SINGLE-TASK MODELS AND THE EXTERNAL BASELINES, ORDERED BY REGION COUNT. CATEGORY IS MACRO-F1, REGION IS ACC@10. BOLD MARKS A STATISTICALLY SUPPORTED IMPROVEMENT OVER THE DEDICATED MODEL; IN THE REGION COLUMNS, $\uparrow$ MARKS THAT SUPPORTED IMPROVEMENT AND $\approx$ A NON-INFERIOR MATCH (TOST, ± 2 PP; SECTION VI-B).

		Next-category (macro-F1)				Next-region (Acc@10)				
Dataset	**Regions**	**Markov-K**	**POI-RGNN**	**Dedicated**	**Joint (ours)**	**HMT-GRN**	**ReHDM**	**STAN**	**Dedicated**	**Joint (ours)**
Istanbul	520	24.55	30.12	54.74 $\pm$0.09	**63.33** $\pm$0.02	60.4	69.33	61.86	75.16 $\pm$0.01	**75.44** $\pm$0.04$^{\uparrow}$
AL	1,109	20.50	23.80	56.82 $\pm$0.03	**64.54** $\pm$0.10	57.05	65.38	60.72	70.11 $\pm$0.10	69.80 $\pm$0.05$^{\approx}$
AZ	1,547	23.92	27.64	56.43 $\pm$0.10	**65.84** $\pm$0.02	43.70	53.00	49.86	59.46 $\pm$0.08	59.56 $\pm$0.05$^{\approx}$
FL	4,703	29.74	34.49	74.51 $\pm$0.03	**79.85** $\pm$0.03	63.74	64.49	72.99	76.70 $\pm$0.02	**77.42** $\pm$0.03$^{\uparrow}$
TX	6,553	28.67	33.03	69.79 $\pm$0.08	**77.23** $\pm$0.12	53.85	48.81‡	61.67†	64.95 $\pm$0.01	**67.07** $\pm$0.45$^{\uparrow}$
CA	8,501	27.58	31.78	70.60 $\pm$0.07	**77.04** $\pm$0.20	49.61	50.26‡	58.52†	63.49 $\pm$0.02	**65.69** $\pm$0.30$^{\uparrow}$

HMT-GRN (region-native, on our folds, scored on the visits whose region appears in training) is the primary external comparison; STAN (faithful, from raw) and ReHDM (its own protocol) are references. Markov-K reports the strongest order per dataset (up to the nine-visit window). †STAN partial-fold cells, truncated by training cost: Texas four of five folds, California two of five (seed 0). ‡ReHDM at Texas and California is a single seed. The joint cells at California and Texas are a single seed over five folds (provisional); every other cell in our columns averages four seeds. $\pm$ is the sd across seeds for four-seed cells, across folds for single-seed cells.

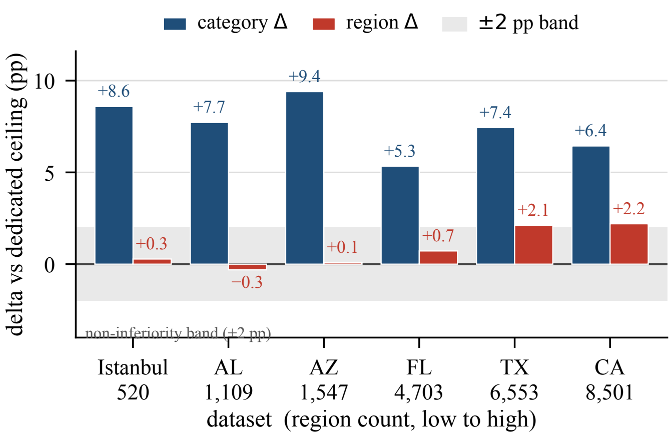

Fig. 4. Signed gains of the joint model over the dedicated single-task models, ordered by region count. The category gain (macro-F1) is positive at every dataset; the region gain (Acc@10) rises with region count across the five U.S. states and is also positive at Istanbul. The two metrics are on different scales; the $\pm$2-point band applies to region.

VII. DISCUSSION AND LIMITATIONS

The clearest reading of our results is a design one: one model is enough. It pairs a shared trunk that carries the semantic context with a private spatial path that keeps next-region accuracy from degrading. The shared context lifts the next-category task sharply over a dedicated model at every dataset, and the private path keeps the next-region task competitive: a non-inferior match (TOST, ± 2 pp) at the two small U.S. states, and a tested gain at the other four datasets. Across the U.S. states the region gain rises with the number of regions, growing rather than fading (Figure 4). Overall, the joint model matches or outperforms the dedicated single-task model on both tasks at once (Table III). For a mobility-aware service, this points to a practical option: one model, one forward pass, can anticipate both what a user will do next and where, without a second model and without a task trade-off.

A short usage sketch makes the motivation concrete. A service could treat the model's top-ranked next regions as an anticipatory set: at California, a shortlist of ten regions out of 8,501 candidates, about a tenth of a percent of the space, contains the true next region 65.69 percent of the time, more than five hundred times better than picking ten at random; at Alabama, ten out of 1,109 capture it 69.80 percent of the time. The predictions are also geographically concentrated. On the four datasets where we measured it (Alabama, Arizona, Florida, and Istanbul; a single seed over five folds), the ten shortlisted regions sit, on average, about 3 to 8 kilometers from the shortlist's own geographic center (the median across visits, computed from region centroids). The miss distance is similar: when the top guess is wrong and the true region was seen in training, the predicted region's centroid sits a median of about 3 to 8 kilometers from the true region's, while two randomly drawn regions from the same map sit a median of 20 to 241 kilometers apart; visits to regions never seen in training are rarer and miss by more. The next category then hints at what to prepare there. This remains motivation, not a measured service result: we report properties of the model's own predictions, not a system's cost or a service's outcome.

Three limits qualify these results. First, the joint numbers at California and Texas come from a single seed over five folds, so those two cells are provisional; the other four datasets average four seeds on both arms. A multi-seed run at the two largest states is the remaining step. Second, our representation is trained once over all places; visits to places never seen during training are the single effect we cannot fully isolate. A planned follow-up study trains the representation on each fold's training places only and extends it to embed unseen visits directly, closing the remaining gap. Third, although a mobility-aware service is our motivation throughout, we do not build or evaluate one; a concrete service evaluation, with its own baselines and costs, remains the natural next study.

Finally, our representation is a fixed per-visit vector, computed once and left unchanged; any downstream model can consume it. Plugging it into other sequence models, and applying it to mobility tasks beyond next-category and next-region, are natural directions to explore. Structural analysis of check-in networks has itself named machine learning as a next step [5]; learning over check-in structure, as this study does, moves in that direction.

VIII. Conclusion

We asked whether one model can predict both the category and the region of a user's next visit. Two findings answer it. First, a check-in-level representation, where each visit gets its own vector instead of one fixed vector per place, makes the next-category task far more learnable than the standard place-level representation does. Second, on top of that representation, a single multi-task model outperforms a dedicated category model at every dataset, and on region it does so at four of the six datasets while remaining non-inferior (TOST, ± 2 pp) at the other two. In summary, one model with a shared semantic context and a private spatial path for region anticipates both what a user will do next and where, without giving up the next-region task. Across the U.S. states the gain on region grows with the number of regions. The code for the model, representation, baselines, and statistical tests is available at https://github.com/VitorHugoOli/PoiMtlNet/tree/mobiwac; both data sources are already public: the category-annotated Gowalla dump (https://figshare.com/articles/dataset/gowalla_data/22126586) and the Massive-STEPS collection [35].

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